**QUESTION 1: [2*5 = 10 points]**

1. Sets A and B have 3 and 4 elements respectively. The total number of possible relations from A to B are ___________? **$2^{12}$**

2. If A = {a, b, c} and relations $R^1, R^2: A \rightarrow A$ are $R^1$ = {(a, b), (b, c), (c, a), (a, a), (b, b), (c, c)} $R^2$ = {(a, b), (b, c), (c, a), (a, a)}, then $R^1$ and $R^2$ are equivalence relations or not? ____**Not**___

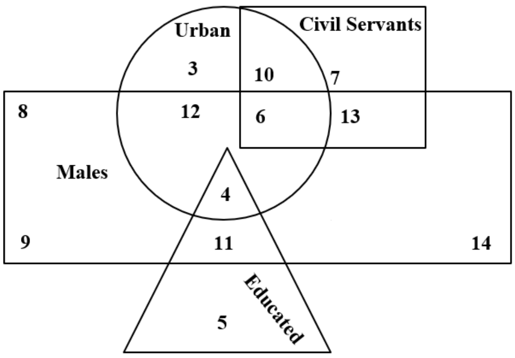
3. $x^2 = xy$ is a relation R, then R is Reflexive, Symmetric, or Transitive? __**Reflexive**___________

4. Let *R* be an equivalence relation on a finite set *A* having *n* elements. Then the number of ordered pairs in *R* is greater than or equal to n? **True**

5. In a class of 100 students, 55 students have passed in Mathematics and 67 students have passed in Physics. Then the number of students who have passed in Physics only is ___________? ___**45**___

**QUESTION 2: Venn diagram:** (A neat and clean Venn diagram is important. Cutting and overwriting can directly affect your marks.) **[ 10 points]**

**A. Refer to the following Venn diagram:** [6]




**I.** Who among the following is a female urban resident and also a civil servant?

**II.** Who among the following is an educated male who hails from urban?

**III.** Who among the following is only a civil servant but neither a male or urban oriented and uneducated?

| | | |
|---|---|---|
| I) 10 | II) 4 | III) 7 |

**B.** During the thorough examination of 70 vehicles at a reputable garage in Chiniot, 15 had tire issues, 20 had brake problems, and 18 exceeded emission limits. Furthermore, 5 vehicles had both tire and brake problems, 6 had issues with both tires and emissions, and 10 had problems with both brakes and emissions. Also, 4 vehicles had faults in all three aspects. How many vehicles passed all assessments without any issues in these three specific areas? Draw an appropriate **Venn Diagram**. [**4**]

**Not Faulty vehicles =?**
**Total vehicles = Faulty vehicles + Not Faulty vehicles**

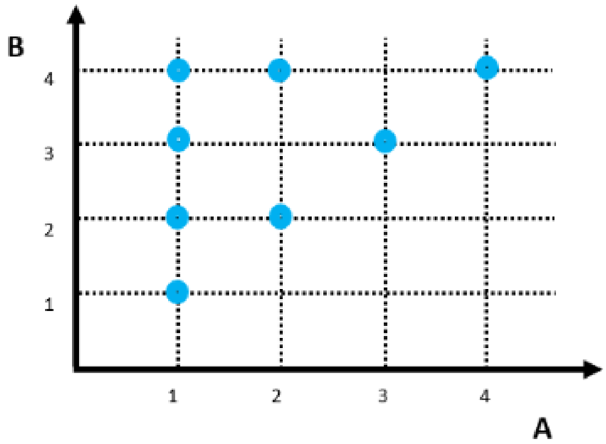
**Not Faulty vehicles = Total – Faulty vehicles**
**Not Faulty vehicles = 70-36**
**Not Faulty vehicles =34**

**QUESTION 3: Solve the following:** Let **A**= {1, 2, 3, 4 } and **B** = {1, 2, 3, 4} . Define relation **R** from set A to set B as follows: **[10 points]**

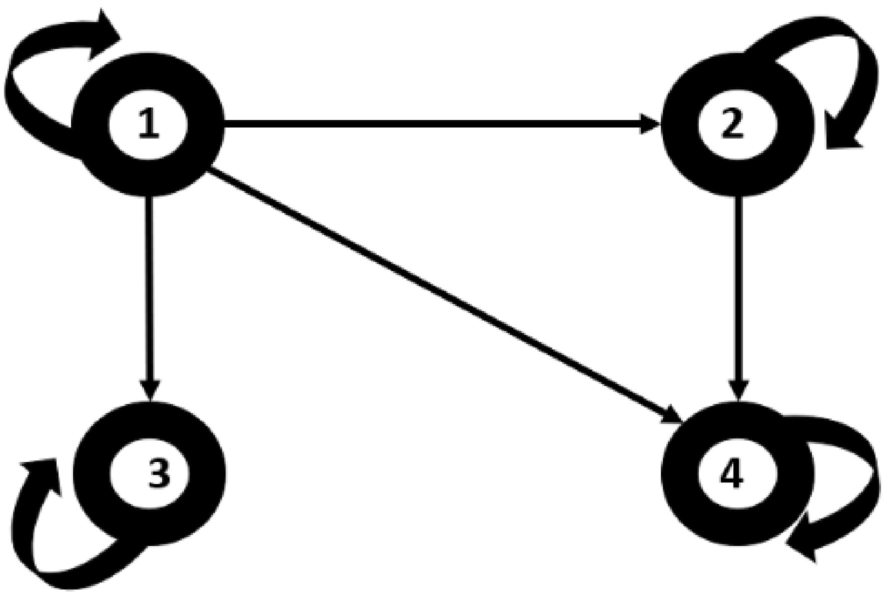
**I. Define a relation R = { (x, y) ∈ A × B | x R y ⇔ x | y}. [4]**

R = { (1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4) }

**II. Find cartesian diagram of a relation R. [3]**




**III. Find directed graph of relation R. [3]**



**QUESTION 4:** **[ 15 points]**

Define a relation '**S**' on a set **'B'** as follows: For any two elements 'x' and 'y' in 'B', 'x S y' if and only if the absolute difference between 'x' and 'y' is less than 5, i.e., |x - y| < 5. Such that **B** belongs to **Z** (Integer Numbers):

- Prove whether the relation 'S' is reflexive. Determine whether, for all 'x' in 'B', 'x S x' holds true.
- Prove whether the relation 'S' is antisymmetric. Demonstrate whether, for all 'x' and 'y' in 'B', if 'x S y' and 'y S x', then 'x' must be equal to 'y' (i.e., 'x = y').
- Prove whether the relation 'S' is transitive. Confirm whether, for all 'x', 'y', and 'z' in 'B', if 'x S y' and 'y S z', then 'x S z' follows.

Based on your **proofs**, determine whether the relation **'S'** is a **partial order** and provide your **reasoning**.

Reflexivity:
For a relation to be reflexive, every element 'x' in 'B' must relate to itself, i.e., 'x S x' must hold true for all 'x' in 'B'.

In this case, if we consider 'x' as any element in 'B', the absolute difference between 'x' and 'x' is always 0 (|x - x| = 0). However, for 'x S x' to be true, the absolute difference should be less than 5. Since 0 is less than 5, 'x S x' is true for all 'x' in 'B'. Therefore, the relation 'S' is reflexive.

Antisymmetry:
A relation is antisymmetric if whenever 'x S y' and 'y S x' hold true, then 'x' must be equal to 'y'.

In this case, if 'x S y' and 'y S x' are both true, it means that both |x - y| < 5 and |y - x| < 5 are satisfied. Since absolute values are non-negative, if |x - y| < 5 and |y - x| < 5 are both true, it implies that 'x' and 'y' are the same. Therefore, the relation 'S' is antisymmetric.

Transitivity:
A relation is transitive if whenever 'x S y' and 'y S z' are true, then 'x S z' must also be true.

Consider 'x S y' and 'y S z'. This means |x - y| < 5 and |y - z| < 5. By the triangle inequality, |x - z| ≤ |x - y| + |y - z|. As both |x - y| < 5 and |y - z| < 5, their sum is less than 10. Hence, |x - z| < 10. Since 10 is greater than 5, it doesn't necessarily guarantee that |x - z| < 5. Therefore, 'x S z' is not always true for all 'x', 'y', and 'z' in 'B'. Thus, the relation 'S' is not transitive.

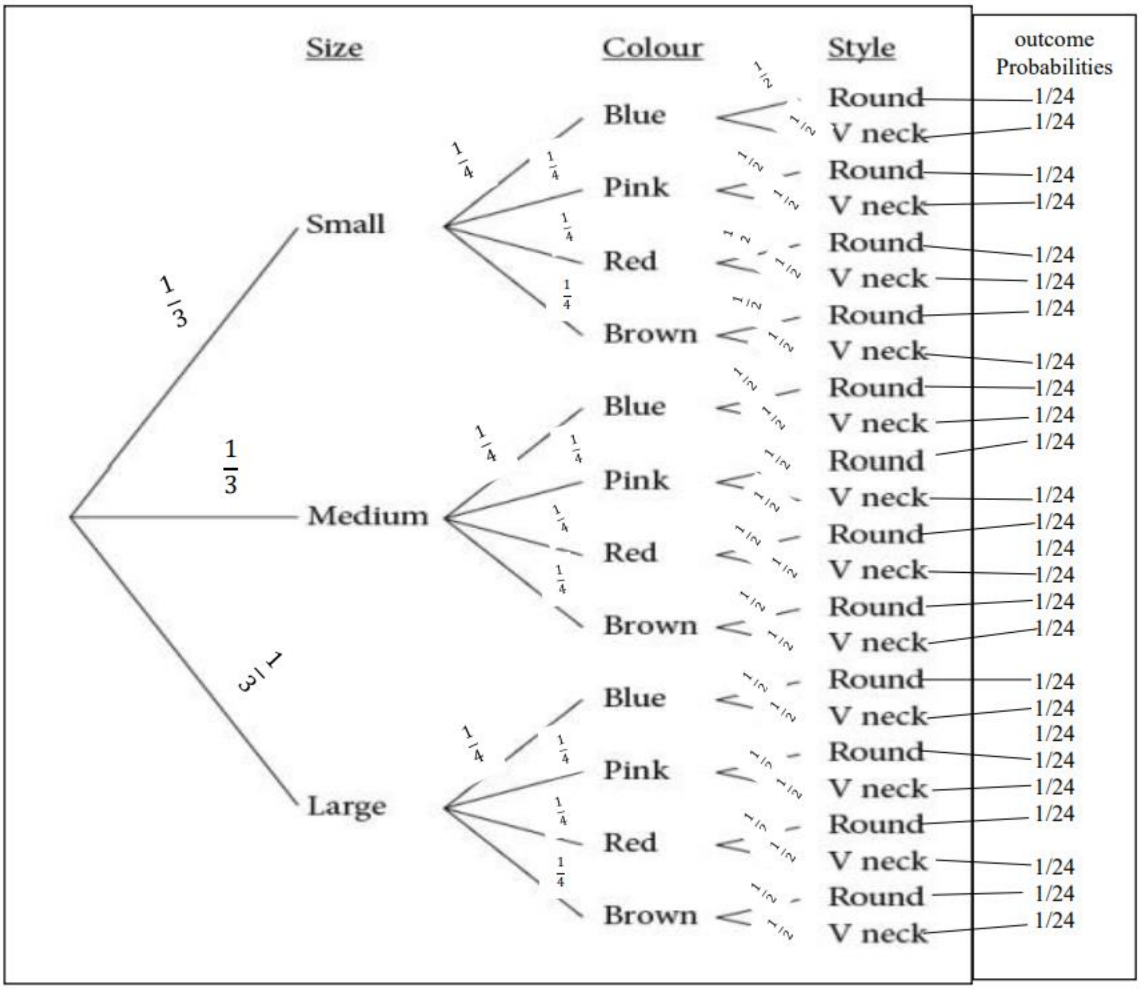
Partial Order:
For a relation to be a partial order, it needs to be reflexive, antisymmetric, and transitive.

'S' is reflexive.
'S' is antisymmetric.
'S' is not transitive.
Since 'S' satisfies reflexivity and antisymmetry but fails in transitivity, it doesn't qualify as a partial order.

**QUESTION 5:** You are buying a jacket from a shop. You have to choose a size, color, and style. You have three choices for size small, medium, and large as labeled (S, M, L), four choices of color for each size blue, pink, red and brown as labeled (BL, PI, RE, BR), and two choices for style in every size and color round neck and V-Neck as labeled (RN, VN).

**[ 10 points]**

1. **Construct the possibility tree showing all possible outcomes/sample of this experiment and compute their event probabilities. [5]**




2. **What is the total number of sample space of this experiment? [2]**

**Sample space =**
**[ S BL RN, S BL VN, S PI RN. S PI VN, S RE RN. S RE VN, S BR VN. S BR RN,**
**M BL RN, M BL VN, M PI RN. M PI VN, M RE RN. M RE VN, M BR VN. M BR RN,**
**L BL RN, L BL VN, L PI RN. L PI VN, L RE RN. L RE VN, L BR VN. L BR RN]**

**Total sample space = 24**

3. **What is the event probability that the jacket you pick doesn't have a round neck? [3]**

$$\textbf{P(does not have neck)} = \frac{P(E)}{P(S)}$$

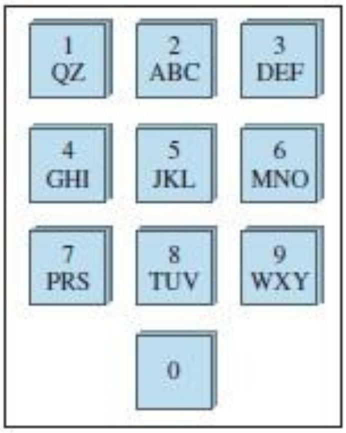
$$= \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24}$$

$$= \frac{1+1+1+1+1+1+1+1+1+1+1+1}{24}$$

$$= \frac{12}{24}$$

$$= \frac{1}{2}$$

**QUESTION 6:** The diagram shows the keypad for an ATM (automatic teller machine). As you can see, the same sequence of keys represents a variety of different PINs. For instance, 2133, AZDE, and BQ3F are all keyed in exactly the same way. **[ 10 points]**




1. **How many different PINs are represented by the same sequence of keys as 5031? [5]**

According to the keypad, the key containing **5** also contains J, K, and L, so we can select the first place in **4 ways**.

The key containing **0** does not contain any other values, so 0 can be selected in **one way**.

Similarly, for the **third** place, there are **4 ways** to select to from 3, D, E, and F.

Finally, for **1**, there are **3 ways** to select, i.e., from 1, Q, Z.

Thus, the number of different pins, representing the same sequence of keys as **5031**, are $4 \times 1 \times 4 \times 3 = 48$.

**2. At an automatic teller machine, each PIN corresponds to a four-digit numeric sequence. For instance, TWJM corresponds to 8956. How many such numeric sequences contain no repeated digit? [5]**

The number of numeric sequences which contain no repeated digits is all of the possible arrangements of the 4 digits from 0 to 9.

Here, 0 can also occur in the first place, as we are not talking about a 4-digit number.

∴The required number of numeric sequences that contain no repeated digits is **P(10, 4)**

$$P(10,4) = \frac{10!}{(10-4)!}$$

$$= \frac{10!}{6!}$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6!}{6!}$$

$$= 10 \times 9 \times 8 \times 7$$

**= 5040**

**GOOD LUCK** 😊